Statement of Purpose

of Abu Noman Md Sakib (Ph.D. applicant for Spring 2025)

Devoting myself to the betterment of the world has always been one of my characteristics. The advancement of computer science, especially artificial intelligence has sparked my curiosity to explore its domains and become a researcher. I grew even more interested when I saw how these technologies have helped every sector make incredible progress in the last few decades. My family taught me the value of education, and I never gave up trying to get the greatest outcome. Being in a developing nation made it difficult to pay for the best education, but it was possible because of the scholarships I was able to obtain throughout my prior academic career. I chose to study in computer science because I have always had a strong passion for technology, computers, and machinery, as well as an increased interest in research.

To fulfill my dream, I got admitted to the Department of Computer Science and Engineering (CSE) at Khulna University of Engineering and Technology (KUET) which is one of the best engineering institutes in Bangladesh. I always kept in mind that the first step to being a good researcher is to have strong problem-solving skills. I was an active member of the programming club on my campus where I had the opportunity to improve my coding skills by learning and implementing different data structures and algorithms. I practiced competitive problem solving in different online platforms and participated in several national and regional programming contests as well. Alongside problem solving, I also built a confectionery management system and library management system where I applied the object oriented features of C++. These projects helped me build a solid foundation for my programming skills. Along with academic studies, I served as Assistant General Secretary (AGS) at the CSE Association of KUET and Organizing Secretary at KUET Adventure Club. Rewarding experiences such as these helped to develop my teamwork and interpersonal skills and the ability to negotiate, resolve conflicts, and empathize with people convincingly.

My first experience in the world of AI came during my internship at Intelligent Machines Limited, an Artificial Intelligence (AI) based software company where I got introduced to the vast field of AI-based applications. I worked closely with senior AI engineers and boosted up my knowledge about large scale AI-systems. My internship experience was beneficial to me when working on AI-related projects and research. One of my projects was on a computer vision based Traffic Signal Violation Detection System from video footage. The backbone of the project was based on YOLOv3 model and the graphical user interface was created using Tkinter. I made this project open source on GitHub and this project has got 431 stars up until now. The satisfaction of developing systems that have real-world applications through my academic projects has fueled my passion for research in computer vision, deep learning, and human-computer interaction. I am eager to explore how these technologies can be integrated to solve practical challenges and drive innovation in real-world scenarios.

During the challenging period of the COVID-19 pandemic, our classes and labs were closed for a long time. At that time, I completely focused on developing my research experience. I worked with a professor from Florida Gulf Coast University. My research works were focused on computer vision, deep learning, and image processing methods for the detection and classification of skin cancer and skin diseases such as eczema and acne. We introduced a new eczema disease dataset and proposed a novel technique for classification. We collected the dataset from hospitals and pre-processed it thoroughly to meet our research needs. After pre-processing, the data was fed into multiple deep neural network models. We also utilized transfer learning to challenge our model performance further. After a lot of effort, our novel architecture successfully outperformed all previous results. Collaborating with international researchers, I successfully contributed to the field by publishing three conference papers. Parallel to the research works, I enrolled in some AI-related online courses on Coursera to gain more in-depth knowledge that might be useful for my research.

In my final year, I completed my thesis on brain hemorrhage classification and segmentation from CT scan images, leveraging my prior knowledge of deep learning and computer vision. A novel lightweight

model for brain hemorrhage classification using Convolutional Neural Networks (CNNs) was introduced. While working with CT scan images, I learned various efficient pre-processing techniques for reducing computational costs and improving model performance. I achieved the highest grade in my academic thesis and was appreciated for my work. The proposed model classifies brain hemorrhage with 96.67% accuracy which was also compared with pre-trained models such as VGG16, VGG19, InceptionV3. This research was published as a book chapter at the International Conference on Information and Communication Technology for Development (ICICTD) 2022. The research was extended by performing segmentation of hemorrhagic images using state-of-the-art models. We applied augmentation techniques for enhancing the dataset which differs from the traditional data augmentation techniques. Twirl and spherical are two non-linear image transformation techniques that were used for the augmentation process. By using U-Net and U-Net++ models, the results on the dataset were validated. This research was published in the International Conference on Sustainable Technologies for Industry (STI) 2022. I also worked on classifying rice and corn leaf diseases using deep learning, with a focus on developing an end-to-end web-based system. In this research, I addressed multiple disease identification challenges from leaf image datasets using CNNs. I built a user-friendly website where users could upload images of leaves, and the system would identify the disease. This research project provided me with the opportunity to learn about the diverse uses of computer vision. Two papers from this research were published in Springer. All of my research works collectively improved my proficiency in literature review, formulating research questions, data analysis, and academic writing.

After my graduation, I started working as a Software Engineer at RedDot Digital Limited, one of the most reputed software companies in Bangladesh. With my 2.5 years of experience as a fullstack developer at RedDot, I have played critical roles in developing high-impact projects such as HR management system and OTT platform for major entities in Bangladesh. This diverse experience has not only sharpened my technical skills but also improved my problem-solving skills. Although I loved my job as a Software Engineer, I felt very limited here because my work required me to use things that others had invented, but there was no innovation on my side. I wanted to create something novel, something that would allow me to create a bigger footprint in the field of Computer Science. This desire is what motivated me to pursue a Ph.D. I am confident that my academic background, research experience, and professional skills will help me to complete the challenging Ph.D. journey successfully. Ultimately, my goal is to build a career as a researcher, contributing to a better world by reducing human effort and enhancing trust in technology.

To complete my vision, I firmly believe the University of Texas at San Antonio will be the best platform as it is one of the top-ranked universities in the United States, with a perfect balance of experimental and theoretical research. The outstanding graduate courses, excellent researchers, and lab facilities persuaded me to apply to this university. I found that the ”San Antonio Virtual Environments” lab of Dr. John Quarles focuses on research in virtual reality, augmented reality and artificial intelligence. I believe my knowledge and his direction will result in excellent outcomes. My prior research work experience will surely contribute to the lab. Dr. Kevin Desai's research on virtual reality, and computer vision also grabbed my attention as his work perfectly coincides with my prior research and publications. Dr. Taslima Akter's research on human computer interaction and accessibility intrigues me. Though my previous works are on deep learning, and computer vision, I feel that my knowledge of artificial intelligence can be helpful in her research lab. Having the opportunity to do research under the supervision of such esteemed academics will provide me with the finest path to achieving my goal of becoming a qualified researcher. I am convinced that earning a Ph.D. from the University of Texas at San Antonio will not only equip me with advanced knowledge and skills but also position me prominently within the research community, which will eventually help me to achieve my career objective.